

# DOMANDE E RISPOSTE MACHINE LEARNING

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Nella **parte 1** si trovano le domande senza highlighting delle risposte corrette, mentre nella **parte 2** si trovano le domande comprendenti delle risposte corrette in **grassetto**.

## PARTE 1: Domande senza risposte corrette

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### 1) Which is different from the others:

- Expectation Maximisation
- Apriori
- K-means
- ☒ Decision Tree

### 2) What is the cross validation:

- ☒ A technique to obtain a good estimation of the performance of a classifier when it will be used with data different from the training set
- A technique to obtain a good estimation of the performance of a classifier with the training set
- ☒ A technique to improve the quality of a classifier
- A technique to improve the speed of a classifier

### 3) What is the single linkage?

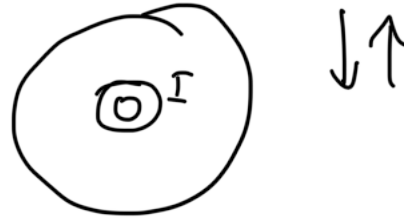
- ☒ A method to compute the distance between two objects, it can be used in hierarchical clustering
- ☒ A method to compute the distance between two sets of items, it can be used in hierarchical clustering
- A method to compute the distance between two classes, it can be used in decision trees
- A method to compute the separation of the objects inside a cluster

### 4) Which of the following characteristic of data can reduce the effectiveness of DBSCAN?

- Clusters have concavities
- All the variables are the same range of values
- Presence of outliers
- ☒ Presence of clusters with different densities

### 5) Which of the following types of data allows the use of the euclidean distance?

- ✓ ~~X~~ Points in a vector space
- Ordered data
  - Transactional data
  - Document representations



### 6) How does pruning work when generating frequent itemsets? ←

- If an itemset is not frequent, then none of its supersets can be frequent, therefore the frequencies of the supersets are not evaluated
- If an itemset is frequent, then none of its supersets can be frequent, therefore the frequencies of the supersets are not evaluated
- ~~no X~~ If an itemset is not frequent, then none of its subsets can be frequent, therefore the frequencies of the subsets are not evaluated
- If an itemset is frequent, then none of its subsets can be frequent, therefore the frequencies of the subsets are not evaluated

### 7) In a decision tree, the number of objects in a node...

- ✓ ~~X~~ ... is smaller than the number of objects in its ancestor
- ... is not related to the number of objects in its ancestor
  - ... is smaller than or equal to the number of objects in its ancestor
  - ... is bigger than the number of objects in its ancestor

### 8) What does K-means try to minimise?

- The distortion, that is the sum of the squared distances of each point with respect to the points of the other clusters
- The distortion, that is the sum of the squared distances of each point with respect to its centroid
- ~~no X~~ The separation, that is the sum of the squared distances of each point with respect to its centroid
- The separation, that is the sum of the squared distances of each cluster centroid with respect to the global centroid of the dataset

### 9) Given the following definitions: TP = True Positives; TN = True Negatives; FP = False Positives; FN = False Negatives. Which of the formulas below computes the precision of a binary classifier?

- TP / (TP + FP)
- ~~no X~~ (TP + TN) / (TP + FP + TN + FN)  $\frac{\text{quante positive vere}}{\text{quante positive trovate}}$
- TN / (TN + FP)
- TP / (TP + FN)

**10) Given the two binary vectors below, which is their similarity according to the Jaccard Coefficient?**

$$\begin{array}{c} 1000101101 \\ 1011101010 \end{array} \quad \frac{3}{8} = 0.375$$

- 0.1
- 0.2
- 0.5
- ✗ 0.375

**11) Consider the transactional dataset below:**

ID Items

1 A,B,C

2 A,B,D

3 B,D,E

4 C,D

5 A,C,D,E

Which is the support of the rule  $A,C \rightarrow B = \frac{n \text{ contains } (A,C,B)}{\text{TOT}} = \frac{1}{5} = 20\%$

- 100%
- ✗ 20%
- 40%
- 50%

**12) Which of the following statements is true?**

- ✓✗ The noise can generate outliers
- The noise always generate outliers
- The data which are similar to the majority are never noise
- ⊙ Outliers can be due to noise

**13) Which of the following clustering methods is not based on distances between objects?**

- Hierarchical Agglomerative
- DBSCAN
- K-Means
- ✓✗ Expectation Maximization

**14) In a decision tree, an attribute which is used only in nodes near the leaves...**

- ... is irrelevant with respect to the target
- ... guarantees high increment of purity
- ⊙ ... gives little insight with respect to the target
- ... has a high correlation with respect to the target

↓  
DETAIL  
ATTRIBUTE

**15) Which of the statements below is true? (Only one)**

NOT EXACT METHOD

- ☒ Sometimes k-means stops to a configuration which does not give the minimum distortion for the chosen value of the number of clusters
- ☐ K-means always stop to a configuration which gives the minimum distortion for the chosen value of the number of clusters
- ☐ K-means works well also with datasets having a very large number of attributes
- ☐ K-means finds the number of clusters which give the minimum distortion

**16) Which of the statements below is true? (One or more)**

- ☒ ☒ Increasing the radius of the neighbourhood can decrease the number of noise points
  - ☐ DBSCAN always stops to a configuration which gives the optimal number of clusters
- ☒ ☒ Sometimes DBSCAN stops to a configuration which does not include any cluster
- ☒ ☒ DBScan can give good performance when clusters have concavities

**17) In data preparation which is the effect of normalisation?**

- ☒ Map all the numeric attributes to the same range, without altering the distribution, in order to avoid that attributes with large ranges have more influence
- ☐ Map all the numeric attributes subtracting the respective mean and dividing by variance
- ☐ Map all the numeric attributes to the same range and to have a Gaussian (or *normal*) *distribution*, in order to avoid that attributes with large ranges have more influence
- ☐ Map all the numeric attributes in order to have a Gaussian (or *normal*) *distribution*, in order to avoid that attributes with large ranges have more influence

**18) Consider the transactional dataset below:**

ID Items

- 1 A,B,C    1 ~
- 2 A,B,D
- 3 B,D,E
- 4 C,D
- 5 A,C,D,E    -

Which is the confidence of the rule  $A,C \rightarrow B$      $= \frac{n \text{ contains } (A,C,B)}{n \text{ contains } (A,C)}$

- ☒ ☒ 50%
- ☐ 40%
- ☐ 20%
- ☐ 100%

### 19) Which of the statements below about Hierarchical Agglomerative Clustering is true?

- Requires the definition of Inertia of clusters
- ✓ ~~X~~ Requires the definition of distance between sets of objects
- Is very efficient also with large datasets
- Is based on a well founded statistical model

### 20) When training a neural network, what is the learning rate?

- ~~NO~~ ~~X~~ The speed of convergence to a stable solution during the learning process
- The ratio between the size of the hidden layer and the input layer of the network
  - The slope of the activation function in a specific node
  - ⊙ A multiplying factor of the correction to be applied to the connection weights

### 21) What are the hyperparameters of a Neural Network? (Possibly non exhaustive)

- Hidden layers structure, ~~Output~~ layer structure, Activation function, Number of epochs
- Network structure, Learning rate, ~~Backpropagation~~ algorithm, Number of epochs
- ~~Input~~ layers structure, Learning rate, Activation function, Number of epochs
- ⊙ Hidden layers structure, Learning rate, Activation function, Number of epochs

### 22) Which of the following preprocessing activities is useful to build a Naive Bayes classifier if the independence hypothesis is violated:

- Discretisation
- Normalisation
- Standardisation
- ✓ ~~X~~ Feature selection

### 23) Which is different from the others?

- ⊙ Silhouette Index → CLUSTERING
- Misclassification Error
- Entropy → CLASSIFICATION
- Gini Index

### 24) Why do we prune a decision tree?

- ⊙ To eliminate parts of the tree where the decisions could be influenced by random effects
- To eliminate rows of the dataset which could be influenced by random effects
- ~~NO~~ ~~X~~ To eliminate parts of the tree where the decision could generate underfitting
- To eliminate attributes which could be influenced by random effects

25) In a dataset with  $D$  attributes, how many subsets of attributes should be considered for feature selection according to an exhaustive search?

- ☒  $O(2^D) \rightarrow$  include/exclude each of  $D$  attributes  
~~☐  $O(D!)$~~   
 •  $O(D)$   
 •  $O(D^2)$

26) In which mining activity the Information Gain can be useful?

- Discovery of association rules
- ☒ Classification
- Clustering
- Discretization

$\hookrightarrow$  FEATURE SELECTION OF ATTRIBUTES THAT BEST DIVIDE THE DATA IN THE NODE

27) Given the two binary vectors below, which is their similarity according to the Simple Matching Coefficient?

|                     |                      |
|---------------------|----------------------|
| 1 0 0 0 1 0 1 1 0 1 | $\frac{5}{10} = 0.5$ |
| 1 1 1 1 1 1 1 1 1 1 |                      |
| 1 0 1 1 1 0 1 0 1 0 |                      |

- 0.1
- 0.3
- ~~☒ 0.5~~
- 0.2

28) Which is the main reason for the standardization of numeric attributes?

- Change the distribution of the numeric attributes, in order to obtain gaussian distributions
- ☒ ~~☐~~ Map all the numeric attributes to a new range such that the mean is zero and the variance is one
- Map all the nominal attributes to the same range, in order to prevent the values with higher frequency from having prevailing influence
- Remove non-standard values

29) Which of the following measure can be used as an alternative to the Information Gain?

- Silhouette Index
- ~~☒~~ Gini Index
- Rand Index
- Jaccard Index

BOTH AIMS TO MEASURE PURITY OF NODE BUT THEY DIFFER ON HOW TO DO IT BY BOTH USING THE PROBABILITY OF A CLASS IN THE SET BUT WITH DIFFERENT FORMULAS

### 30) Which of the following is not an objective of feature selection

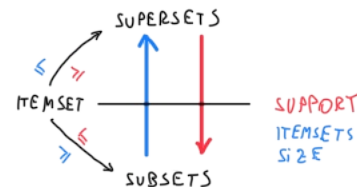
- ⊙ Select the features with higher range, which have more influence on the computation
- Reduce time and memory complexity of the mining algorithms
- Reduce the effect of noise
- Avoid the curse of dimensionality

### 31) After fitting DBSCAN with the default parameter values the results are: 0 clusters, 100% of noise points. Which will be your next trial?

- ✓ ✗ Reduce the minimum number of objects in the neighborhood
  - Reduce the minimum number of objects in the neighborhood and the radius of the neighborhood
- ✓ ✗ Increase the radius of the neighborhood
  - Decrease the radius of the neighborhood

### 32) What is the meaning of the statement "the support is anti-monotone"?

- ✓ ✗ The support of an itemset never exceeds the support if its subsets
  - The support of an itemset is always smaller than the support of its supersets
  - The support of an itemsets never exceeds the support if its supersets
  - The support of an itemset is always smaller than the support of its subsets



### 33) In feature selection, what is the Principal Component Analysis?

- ✓ • A mathematical technique used to transform non numeric attributes into numeric attributes
- ✓ ✗ A mathematical technique used to transform a set of numeric attributes into a smaller set of numeric attributes which capture most of the variability in data
- A heuristic technique used to find a subset of the attributes which produces the same classifier
- A mathematical technique used to find the principal attributes which determine the classification process

### 34) A Decision Tree is...

- A tree-structured plan of tests on single attributes to forecast the cluster
- ⊙ A tree-structured plan of tests on single attributes to forecast the target → REAL AIM
- NO ✗ A tree-structured plan of tests on single attributes to obtain the maximum purity of a node only a consequence
- A tree-structured plan of tests on multiple attributes to forecast the target

**35) In data preprocessing, which of the following are the objectives of the aggregation of attributes?**

- Obtain a more detailed description of data
- ✓ ~~X~~ Obtain a less detailed scale
- ✓ ~~X~~ Reduce the variability of data
- ✓ ~~X~~ Reduce the number of attributes or distinct values

**36) In order to reduce the dimensionality of a dataset, which is the advantage of Multi Dimensional Scaling (MDS), with respect to Principal Component Analysis (PCA)**

- MDS requires less computational power
- ⊙ MDS can be used also with categorical data, provided that the matrix of the distance is available, while PCA is limited to vector spaces
- MDS can be used with categorical data after a transformation in a vector space
- NO ~~X~~ MDS can work on any kind of data, while PCA is limited to categorical data

**37) What is the main purpose of smoothing in Bayesian classification?**

- ⊙ Classifying an object containing attribute values which are missing from some classes in the training set → in order to deal with 0 probabilities
- Dealing with missing values
- NO ~~X~~ Reduce the variability of the data → EFFECT
- Classifying an object containing attribute values which are missing from some classes in the test set

**38) Consider the transactional dataset below:**

ID Items

1 A,B,C

2 A,B,D

3 B,D,E

4 C,D

5 A,C,D,E

**Which is the confidence of the rule B -> E?** 
$$= \frac{B \wedge E}{B} = \frac{1}{3} = 0.33$$

- 100%
- 50%
- ✓ ~~X~~ 33%
- 20%



### 39) In a Decision Tree for classification, what is a leaf node?

- ⊙ A node which assigns a class value to the objects passing the tests on the path from the root to the node itself
- ~~NO~~ X A node where all the objects belong to the same class → NOT EXACT
  - A node which allows classification without errors
  - A node which assigns a class value only by majority of examples

### 40) Match the rule evaluation formulas with their names:

1.  $\text{sup}(A \cup C) - \text{sup}(A)\text{sup}(C)$
2.  $\frac{\text{conf}(A \Rightarrow C)}{\text{sup}(C)}$
3.  $\frac{\text{sup}(A \Rightarrow C)}{\text{sup}(A)}$
4.  $\frac{1 - \text{sup}(C)}{1 - \text{conf}(A \Rightarrow C)}$

Options:

- Confidence → 3
- Lift → 2 → confidence with opposite denominator
- Leverage → 1 → 1-level formula
- Conviction → 4

### 41) Which of the statements below best describes the strategy of Apriori in finding the frequent itemsets?

- Evaluation of the support of the itemsets in an order such that uninteresting parts of the search space are considered only at the end of the execution
- Evaluation of the confidence of the itemsets in an order such that uninteresting parts of the search space are pruned as soon as possible
- ✓ X Evaluation of the support of the itemsets in an order such that uninteresting parts of the search space are pruned as soon as possible
- Evaluation of the support of the itemsets in an order such that the interesting parts of the search space are pruned as soon as possible

### 42) What is the Gini index?

- A measure of the entropy of a dataset
- An accuracy measure of a dataset alternative to the Information Gain and to the Misclassification Index
- ✓ X An impurity measure of a dataset alternative to the Information Gain and to the Misclassification Index
- An impurity measure of a dataset alternative to overfitting and underfitting

### 43) What measure is maximised by the Expectation Maximisation algorithm for clustering?

- ✓✗ The likelihood of a class label, given the attributes of the example
- The likelihood of an attribute, given the class label
- The likelihood of an example
- The support of a class

### 44) Given the following definitions: TP = True Positives; TN = True Negatives; FP = False Positives; FN = False Negatives. Which of the formulas below computes the accuracy of a binary classifier?

- $TN / (TN + FP)$
- ✓✗  $(TP + TN) / (TP + FP + TN + FN)$
- $TP / (TP + FN)$
- $TP / (TP + FP)$

### 45) In data preprocessing, which of the following is not an objective of the aggregation of attributes

- Reduce the variability of data
- Obtain a less detailed scale
- Reduce the number of attributes or objects
- ✓✗ Obtain a more detailed description of data

### 46) Which of the following is a strength of the clustering algorithm DBSCAN?

- Requires to set the number of clusters as a parameter
- ✓✗ Ability to find cluster with concavities
- ✓✗ Ability to separate outliers from regular data
- Very fast by computation

### 47) Which of the following statements regarding the discovery of association rules is true? (One or more)

- ✓✗ The confidence of a rule can be computed starting from the supports of the itemsets
- ✓✗ The support of an itemset is anti-monotonic with respect to the composition of the itemset
- The support of a rule can be computed given the confidence of the rule
- The confidence of an itemset is anti-monotonic with respect to the composition of the itemset

### 48) For each type of data choose the best suited distance function

- Vector space with real values -> **EUCLIDEAN DISTANCE**
- Vectors of terms representing documents -> **COSINE DISTANCE** → ANGLES BETWEEN FREQUENCY VECTORS OF TEXTS
- High dimensional spaces -> **MANHATTAN DISTANCE** → AVOIDS CURSE OF DIMENSIONALITY
- Boolean data -> **JACCARD COEFFICIENT** → SUITED FOR BINARY VECTORS

**49) Which of the following is not a strength point of Dbscan with respect to K-means**

- The robustness with respect to outliers
- The effectiveness, even in presence of noise
- The effectiveness even if there are clusters with non-convex shape
- ✓✗ The efficiency even in large datasets

**50) Which is the effect of the curse of dimensionality**

- When the number of dimensions increases the results tend to be prone to overfitting
- ⦿ When the number of dimensions increases the euclidean distance becomes less effective to discriminate between points in the space
- NO✗ When the number of dimensions increases the computing power necessary to compute the distances becomes too high
- When the number of dimensions increases the classifiers cannot be correctly tuned

**51) Which is different from the others?**

- Expectation Maximisation
- ✓✗ Decision Tree
- K-means
- Dbscan

**52) In a Neural Network, what is the backpropagation?**

- NO✗ The technique used to adjust the ~~node~~ weights according to the difference between the desired output and the output generated by the network
- The technique used to adjust the weights limiting the probability of overfitting
- The technique used to adjust the output according to the difference between the desired weights and the actual weights
- ⦿ The technique used to adjust the connection weights according to the difference between the desired output and the output generated by the network

**53) Which is different from the others?**

- ✓✗ Dbscan
- SVM
- Neural Network
- Decision Tree

**54) Which of the following is not a property of a metric distance function**

- Symmetry
- Positive definiteness
- Triangle inequality
- ✓✗ Boundedness

**55) Which of the statements below is true? (One or more)**

- K-mean always stops to a configuration which gives the minimum distortion for the chosen value of the number of clusters
- ✓✗ K-means is very sensitive to the initial assignment of the centers
- ⊙ K-means is quite efficient even for large datasets → LINEAR COMPLEXITY AND SIMPLE
- ✓✗ Sometimes k-means stops to a configuration which does not give the minimum distortion for the chosen value of the number of clusters

**56) Which of the following characteristic of data can reduce the effectiveness of K-Means?**

- Presence of values with high frequency
- ✓✗ Presence of outliers
- All the variables have the same distribution of values
- NO ✗ All the variables have the same range of values SOMO STUPIDO

**57) Which is the purpose of discretisation?**

- Increase the number of distinct values in an attribute, in order to put in evidence possible patterns and regularities
- Reduce the number of distinct values in an attribute, in order to increase the efficiency of the computations
- ✓✗ Reduce the number of distinct values in an attribute, in order to put in evidence possible patterns and regularities
- Reduce the range of values of a numeric attribute, to make all the attributes more comparable

**58) Given the following definitions: TP = True Positives; TN = True Negatives; FP = False Positives; FN = False Negatives. Which of the formulas below computes the recall of a binary classifier?**

- ⊙  $TP / (TP + FN)$
- NO ✗  $TN / (TN + FP)$
- $TP / (TP + FP)$
- $(TP + TN) / (TP + FP + TN + FN)$

**59) The Information Gain is used to**

- ✓✗ select the attribute which maximises, for a given training set, the ability to predict the class value
- select the attribute which maximises, for a given test set, the ability to predict the class value
- select the attribute which maximises, for a given training set, the ability to predict all the other attribute values
- select the class with maximum probability

**60) Which is the main reason for the MinMax scaling (also known as "rescaling") of attributes?**

- Change the distribution of the numeric attributes, in order to obtain gaussian distributions
- ☒ ~~NO~~ Map all the ~~nominal~~ attributes to the same range in order to prevent the values with higher frequency from having prevailing influence
- Remove abnormal values
- ☒ Map all the numeric attributes to the same range, in order to prevent attributes with higher range from having prevalent influence

**61) Which of the following is a base hypothesis for a bayesian classifier?**

- ☒ The attributes must be statistically independent inside each class
- The attributes must have negative correlation
- ☒ ~~NO~~ The attributes must be statistically independent
- The attributes must have zero correlation

**62) When developing a classifier, which of the following is a symptom of overfitting?**

- The error rate in the test set is much smaller than the error rate in the training set
- The precision is much greater than the recall
- The error rate in the test set is more than 30%
- ☒ ~~NO~~ The error rate in the test set is much greater than the error rate in the training set

**63) With reference to the total sum of squared errors and separation of a clustering scheme, which of the statements below is true?**

- They are two ways to measure the same thing
- They are strictly correlated, if, changing the clustering scheme, one increases, then the other does the same
- It is possible to optimise them (i.e. minimise SSE and maximise SSB) separately
- ☒ ~~NO~~ They are strictly correlated, if, changing the clustering scheme, one increases, then the other decreases

## 64) How can we measure the quality of a trained regression model?

- ⊙ With a formula elaborating the difference between the forecast values and the true ones
- NO ✗ With a confusion matrix → **USED IN CLASSIFICATION!** → IN REGRESSION WE DEAL WITH CONTINUOUS VALUES WHILE IN CLASSIFICATION WITH DISCRETE VALUES
  - Counting the number of values correctly forecast
  - With precision, recall and accuracy

## 65) What is the coefficient of determination R<sup>2</sup>?

- ✓ ✗ Provide an index of goodness for a linear regression model
  - Measure the amount of error in a linear regression model
  - Measure the amount of error in a regression model
  - An index of goodness for a classification model

## 66) What measure is maximised by the Expectation Maximisation algorithm for clustering?

Scegli un'alternativa:

- ✓ ✗ The likelihood the distributions, defined by the parameters found, given the data available
  - The likelihood of an example
  - The support of a class
  - The likelihood of an attribute, given the class label

## 67) What is the difference between classification and regression?

- ⊙ Classification has a categorical target, while regression has a numeric target
- NO ✗ Classification is a supervised activity. while regression is unsupervised **BOTH SUPERVISED !!**
  - Classification can make errors, while regression is always exact
  - Classification can have a numeric or categorical target, while regression has always a categorical target

## 68) When is polynomial regression appropriate?

- ✓ ✗ When the relationship between the predicting variable and the target cannot be approximated as linear
  - When the target values are not linearly separable
  - When there is more than one predicting attribute
  - When it is necessary to project the data into a dimensional space

## 69) How is the number of clusters determined in agglomerative clustering?

- By assigning each data point to a cluster based on the final structure obtained from the dendrogram.
- ✓ ✗ By cutting the dendrogram at a certain height.
  - By merging the closest clusters according to the defined distance metric.
  - By computing the pairwise similarity or distance between all clusters.

## 70) In data preprocessing, which of the operations below performs aggregation?

- No** ✗ Combination of two attributes in order to obtain a more general attribute with smaller variability
- ☒ Reduction of the number of distinct values of an attribute, in order to reduce its variability
- ☒ Reduction of the number of rows of the dataset by applying a grouping on one or more attributes and computing the measures as the results of aggregation functions
- Combination of two or more datasets in order to obtain more information

## 71) Match specific situations of data to the most appropriate pre-processing activity

- The numeric attributes have very diverse ranges of values : 3 ✓
- An attribute has a very skewed distribution (e.g. most of the values concentrated in a small range, and a very long tail of outliers): 4 ✓
- A categorical attribute has a too large number of distinct values: 1 ✓
- A nominal attribute needs to be used as predicting when using the sklearn library: 5 ✓

Alternatives:

- 1 • Aggregation
- 2 • Ordinal Encoding
- 3 • Rescaling
- 4 • Non linear transformations
- 5 • OneHot Encoding

## 72) What is the first step in agglomerative clustering?

Scegli un'alternativa:

- Cutting the Dendrogram: Decide on the number of clusters by cutting the dendrogram at a certain height
- Pairwise Cluster Similarity Calculation: Compute the pairwise similarity or distance between all clusters
- ✓** ✗ Initialization: Start with each data point as a singleton cluster
- Cluster Assignment: Assign each data point to a cluster based on the final structure obtained from the dendrogram

## 73) Which of the following situations could suggest the use of MinMax scaling? (also known as "rescaling")

- ☒ There are numeric attributes with very diverse value ranges
- There are attributes with very skewed data distributions
  - There are attributes with a very large number of missing values
- No** ✗ There is a numeric attribute with values spread in a very large range

## PARTE 2: Domande con risposte corrette (in grassetto)

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### 1) Which is different from the others:

- Expectation Maximisation
- Apriori
- K-means
- **Decision Tree**

### 2) What is the cross validation:

- **A technique to obtain a good estimation of the performance of a classifier when it will be used with data different from the training set**
- A technique to obtain a good estimation of the performance of a classifier with the training set
- A technique to improve the quality of a classifier
- A technique to improve the speed of a classifier

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- **A method to compute the distance between two sets of items, it can be used in hierarchical clustering**
- A method to compute the distance between two classes, it can be used in decision trees
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- **Presence of clusters with different densities**

### 5) Which of the following types of data allows the use of the euclidean distance?

- **Points in a vector space**
- Ordered data
- Transactional data
- Document representations



## 6) How does pruning work when generating frequent itemsets?

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## 7) In a decision tree, the number of objects in a node...

- **... is smaller than the number of objects in its ancestor**
- ... is not related to the number of objects in its ancestor
- ... is smaller than or equal to the number of objects in its ancestor
- ... is bigger than the number of objects in its ancestor

## 8) What does K-means try to minimise?

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- **The distortion, that is the sum of the squared distances of each point with respect to its centroid**
- The separation, that is the sum of the squared distances of each point with respect to its centroid
- The separation, that is the sum of the squared distances of each cluster centroid with respect to the global centroid of the dataset

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- $TN / (TN + FP)$
- $TP / (TP + FN)$

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1000101101

1011101010

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- 0.2
- 0.5
- **0.375**

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- **20%**
- 40%
- 50%

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- The noise always generate outliers
- The data which are similar to the majority are never noise
- **Outliers can be due to noise**

**13) Which of the following clustering methods is not based on distances between objects?**

- Hierarchical Agglomerative
- DBSCAN
- K-Means
- **Expectation Maximization**

**14) In a decision tree, an attribute which is used only in nodes near the leaves...**

- ... is irrelevant with respect to the target
- ... guarantees high increment of purity
- **... gives little insight with respect to the target**
- ... has a high correlation with respect to the target

**15) Which of the statements below is true? (Only one)**

- **Sometimes k-means stops to a configuration which does not give the minimum distortion for the chosen value of the number of clusters**
- K-means always stop to a configuration which gives the minimum distortion for the chosen value of the number of clusters
- K-means works well also with datasets having a very large number of attributes
- K-means finds the number of clusters which give the minimum distortion

**16) Which of the statements below is true? (One or more)**

- **Increasing the radius of the neighbourhood can decrease the number of noise points**
- DBSCAN always stops to a configuration which gives the optimal number of clusters
- **Sometimes DBSCAN stops to a configuration which does not include any cluster**
- **DBScan can give good performance when clusters have concavities**

**17) In data preparation which is the effect of normalisation?**

- **Map all the numeric attributes to the same range, without altering the distribution, in order to avoid that attributes with large ranges have more influence**
- Map all the numeric attributes subtracting the respective mean and dividing by variance
- Map all the numeric attributes to the same range and to have a Gaussian (or *normal*) *distribution*, in order to avoid that attributes with large ranges have more influence
- Map all the numeric attributes in order to have a Gaussian (or *normal*) *distribution*, in order to avoid that attributes with large ranges have more influence

**18) Consider the transactional dataset below:**

ID Items

1 A,B,C

2 A,B,D

3 B,D,E

4 C,D

5 A,C,D,E

**Which is the confidence of the rule  $A,C \rightarrow B$**

- **50%**
- 40%
- 20%
- 100%

**19) Which of the statements below about Hierarchical Agglomerative Clustering is true?**

- Requires the definition of Inertia of clusters
- **Requires the definition of distance between sets of objects**
- Is very efficient also with large datasets
- Is based on a well founded statistical model

**20) When training a neural network, what is the learning rate?**

- The speed of convergence to a stable solution during the learning process
- The ratio between the size of the hidden layer and the input layer of the network
- The slope of the activation function in a specific node
- **A multiplying factor of the correction to be applied to the connection weights**

**21) What are the hyperparameters of a Neural Network? (Possibly non exhaustive)**

- Hidden layers structure, Output layer structure, Activation function, Number of epochs
- Network structure, Learning rate, Backpropagation algorithm, Number of epochs
- Input layers structure, Learning rate, Activation function, Number of epochs
- **Hidden layers structure, Learning rate, Activation function, Number of epochs**

**22) Which of the following preprocessing activities is useful to build a Naive Bayes classifier if the independence hypothesis is violated:**

- Discretisation
- Normalisation
- Standardisation
- **Feature selection**

**23) Which is different from the others?**

- **Silhouette Index**
- Misclassification Error
- Entropy
- Gini Index

**24) Why do we prune a decision tree?**

- **To eliminate parts of the tree where the decisions could be influenced by random effects**
- To eliminate rows of the dataset which could be influenced by random effects
- To eliminate parts of the tree where the decision could generate underfitting
- To eliminate attributes which could be influenced by random effects

**25) In a dataset with  $D$  attributes, how many subsets of attributes should be considered for feature selection according to an exhaustive search?**

- **$O(2^D)$**
- $O(D!)$
- $O(D)$
- $O(D^2)$

**26) In which mining activity the Information Gain can be useful?**

- Discovery of association rules
- **Classification**
- Clustering
- Discretization

**27) Given the two binary vectors below, which is their similarity according to the Simple Matching Coefficient?**

1 0 0 0 1 0 1 1 0 1  
1 0 1 1 1 0 1 0 1 0

- 0.1
- 0.3
- **0.5**
- 0.2

**28) Which is the main reason for the standardization of numeric attributes?**

- Change the distribution of the numeric attributes, in order to obtain gaussian distributions
- **Map all the numeric attributes to a new range such that the mean is zero and the variance is one**
- Map all the nominal attributes to the same range, in order to prevent the values with higher frequency from having prevailing influence
- Remove non-standard values

**29) Which of the following measure can be used as an alternative to the Information Gain?**

- Silhouette Index
- **Gini Index**
- Rand Index
- Jaccard Index

**30) Which of the following is not an objective of feature selection**

- **Select the features with higher range, which have more influence on the computation**
- Reduce time and memory complexity of the mining algorithms
- Reduce the effect of noise
- Avoid the curse of dimensionality

**31) After fitting DBSCAN with the default parameter values the results are: 0 clusters, 100% of noise points. Which will be your next trial?**

- **Reduce the minimum number of objects in the neighborhood**
- Reduce the minimum number of objects in the neighborhood and the radius of the neighborhood
- **Increase the radius of the neighborhood**
- Decrease the radius of the neighborhood

**32) What is the meaning of the statement "*the support is anti-monotone*"?**

- **The support of an itemset never exceeds the support of its subsets**
- The support of an itemset is always smaller than the support of its supersets
- The support of an itemsets never exceeds the support of its supersets
- The support of an itemset is always smaller than the support of its subsets

**33) In feature selection, what is the Principal Component Analysis?**

- A mathematical technique used to transform non numeric attributes into numeric attributes
- **A mathematical technique used to transform a set of numeric attributes into a smaller set of numeric attributes which capture most of the variability in data**
- A heuristic technique used to find a subset of the attributes which produces the same classifier
- A mathematical technique used to find the principal attributes which determine the classification process

**34) A Decision Tree is...**

- A tree-structured plan of tests on single attributes to forecast the cluster
- **A tree-structured plan of tests on single attributes to forecast the target**
- A tree-structured plan of tests on single attributes to obtain the maximum purity of a node
- A tree-structured plan of tests on multiple attributes to forecast the target

**35) In data preprocessing, which of the following are the objectives of the aggregation of attributes?**

- Obtain a more detailed description of data
- **Obtain a less detailed scale**
- **Reduce the variability of data**
- **Reduce the number of attributes or distinct values**

**36) In order to reduce the dimensionality of a dataset, which is the advantage of Multi Dimensional Scaling (MDS), with respect to Principal Component Analysis (PCA)**

- MDS requires less computational power
- **MDS can be used also with categorical data, provided that the matrix of the distance is available, while PCA is limited to vector spaces**
- MDS can be used with categorical data after a transformation in a vector space
- MDS can work on any kind of data, while PCA is limited to categorical data

**37) What is the main purpose of smoothing in Bayesian classification?**

- **Classifying an object containing attribute values which are missing from some classes in the training set**
- Dealing with missing values
- Reduce the variability of the data
- Classifying an object containing attribute values which are missing from some classes in the test set

**38) Consider the transactional dataset below:**

ID Items

1 A,B,C

2 A,B,D

3 B,D,E

4 C,D

5 A,C,D,E

**Which is the confidence of the rule B -> E ?**

- 100%
- 50%
- **33%**
- 20%

**39) In a Decision Tree for classification, what is a leaf node?**

- **A node which assigns a class value to the objects passing the tests on the path from the root to the node itself**
- A node where all the objects belong to the same class
- A node which allows classification without errors
- A node which assigns a class value only by majority of examples

**40) Match the rule evaluation formulas with their names:**

1.  $\text{sup}(A \cup C) - \text{sup}(A)\text{sup}(C)$

2.  $\frac{\text{conf}(A \Rightarrow C)}{\text{sup}(C)}$

3.  $\frac{\text{sup}(A \Rightarrow C)}{\text{sup}(A)}$

4.  $\frac{1 - \text{sup}(C)}{1 - \text{conf}(A \Rightarrow C)}$

Options:

- Confidence → 3
- Lift → 2
- Leverage → 1
- Conviction → 4



**41) Which of the statements below best describes the strategy of Apriori in finding the frequent itemsets?**

- Evaluation of the support of the itemsets in an order such that uninteresting parts of the search space are considered only at the end of the execution
- Evaluation of the confidence of the itemsets in an order such that uninteresting parts of the search space are pruned as soon as possible
- **Evaluation of the support of the itemsets in an order such that uninteresting parts of the search space are pruned as soon as possible**
- Evaluation of the support of the itemsets in an order such that the interesting parts of the search space are pruned as soon as possible

**42) What is the Gini index?**

- A measure of the entropy of a dataset
- An accuracy measure of a dataset alternative to the Information Gain and to the Misclassification Index
- **An impurity measure of a dataset alternative to the Information Gain and to the Misclassification Index**
- An impurity measure of a dataset alternative to overfitting and underfitting

**43) What measure is maximised by the Expectation Maximisation algorithm for clustering?**

- **The likelihood of a class label, given the attributes of the example**
- The likelihood of an attribute, given the class label
- The likelihood of an example
- The support of a class

**44) Given the following definitions: TP = True Positives; TN = True Negatives; FP = False Positives; FN = False Negatives. Which of the formulas below computes the accuracy of a binary classifier?**

- $TN / (TN + FP)$
- **$(TP + TN) / (TP + FP + TN + FN)$**
- $TP / (TP + FN)$
- $TP / (TP + FP)$

**45) In data preprocessing, which of the following is not an objective of the aggregation of attributes**

- Reduce the variability of data
- Obtain a less detailed scale
- Reduce the number of attributes or objects
- **Obtain a more detailed description of data**

**46) Which of the following is a strength of the clustering algorithm DBSCAN?**

- Requires to set the number of clusters as a parameter
- **Ability to find cluster with concavities**
- **Ability to separate outliers from regular data**
- Very fast by computation

**47) Which of the following statements regarding the discovery of association rules is true? (One or more)**

- **The confidence of a rule can be computed starting from the supports of the itemsets**
- **The support of an itemset is anti-monotonic with respect to the composition of the itemset**
- The support of a rule can be computed given the confidence of the rule
- The confidence of an itemset is anti-monotonic with respect to the composition of the itemset

**48) For each type of data choose the best suited distance function**

- Vector space with real values -> **euclidean distance**
- Vectors of terms representing documents -> **cosine distance**
- High dimensional spaces -> **manhattan distance**
- Boolean data -> **jaccard coefficient**

**49) Which of the following is not a strength point of Dbscan with respect to K-means**

- The robustness with respect to outliers
- The effectiveness, even in presence of noise
- The effectiveness even if there are clusters with non-convex shape
- **The efficiency even in large datasets**

**50) Which is the effect of the curse of dimensionality**

- When the number of dimensions increases the results tend to be prone to overfitting
- **When the number of dimensions increases the euclidean distance becomes less effective to discriminate between points in the space**
- When the number of dimensions increases the computing power necessary to compute the distances becomes too high
- When the number of dimensions increases the classifiers cannot be correctly tuned

**51) Which is different from the others?**

- Expectation Maximisation
- **Decision Tree**
- K-means
- Dbscan

**52) In a Neural Network, what is the backpropagation?**

- The technique used to adjust the node weights according to the difference between the desired output and the output generated by the network
- The technique used to adjust the weights limiting the probability of overfitting
- The technique used to adjust the output according to the difference between the desired weights and the actual weights
- **The technique used to adjust the connection weights according to the difference between the desired output and the output generated by the network**

**53) Which is different from the others?**

- **Dbscan**
- SVM
- Neural Network
- Decision Tree

**54) Which of the following is not a property of a metric distance function**

- Symmetry
- Positive definiteness
- Triangle inequality
- **Boundedness**

**55) Which of the statements below is true? (One or more)**

- K-mean always stops to a configuration which gives the minimum distortion for the chosen value of the number of clusters
- **K-means is very sensitive to the initial assignment of the centers**
- **K-means is quite efficient even for large datasets**
- **Sometimes k-means stops to a configuration which does not give the minimum distortion for the chosen value of the number of clusters**

**56) Which of the following characteristic of data can reduce the effectiveness of K-Means?**

- Presence of values with high frequency
- **Presence of outliers**
- All the variables have the same distribution of values
- All the variables have the same range of values

**57) Which is the purpose of discretisation?**

- Increase the number of distinct values in an attribute, in order to put in evidence possible patterns and regularities
- Reduce the number of distinct values in an attribute, in order to increase the efficiency of the computations
- **Reduce the number of distinct values in an attribute, in order to put in evidence possible patterns and regularities**
- Reduce the range of values of a numeric attribute, to make all the attributes more comparable

**58) Given the following definitions: TP = True Positives; TN = True Negatives; FP = False Positives; FN = False Negatives. Which of the formulas below computes the recall of a binary classifier?**

- **$TP / (TP + FN)$**
- $TN / (TN + FP)$
- $TP / (TP + FP)$
- $(TP + TN) / (TP + FP + TN + FN)$

**59) The Information Gain is used to**

- **select the attribute which maximises, for a given training set, the ability to predict the class value**
- select the attribute which maximises, for a given test set, the ability to predict the class value
- select the attribute which maximises, for a given training set, the ability to predict all the other attribute values
- select the class with maximum probability

**60) Which is the main reason for the MinMax scaling (also known as "rescaling") of attributes?**

- Change the distribution of the numeric attributes, in order to obtain gaussian distributions
- Map all the nominal attributes to the same range in order to prevent the values with higher frequency from having prevailing influence
- Remove abnormal values
- **Map all the numeric attributes to the same range, in order to prevent attributes with higher range from having prevalent influence**

**61) Which of the following is a base hypothesis for a bayesian classifier?**

- **The attributes must be statistically independent inside each class**
- The attributes must have negative correlation
- The attributes must be statistically independent
- The attributes must have zero correlation

**62) When developing a classifier, which of the following is a symptom of overfitting?**

- The error rate in the test set is much smaller than the error rate in the training set
- The precision is much greater than the recall
- The error rate in the test set is more than 30%
- **The error rate in the test set is much greater than the error rate in the training set**

**63) With reference to the total sum of squared errors and separation of a clustering scheme, which of the statements below is true?**

- They are two ways to measure the same thing
- They are strictly correlated, if, changing the clustering scheme, one increases, then the other does the same
- It is possible to optimise them (i.e. minimise SSE and maximise SSB) separately
- **They are strictly correlated, if, changing the clustering scheme, one increases, then the other decreases**

**64) How can we measure the quality of a trained regression model?**

- **With a formula elaborating the difference between the forecast values and the true ones**
- With a confusion matrix
- Counting the number of values correctly forecast
- With precision, recall and accuracy

**65) What is the coefficient of determination  $R^2$ ?**

- **Provide an index of goodness for a linear regression model**
- Measure the amount of error in a linear regression model
- Measure the amount of error in a regression model
- An index of goodness for a classification model

**66) What measure is maximised by the Expectation Maximisation algorithm for clustering?**

Scegli un'alternativa:

- **The likelihood the distributions, defined by the parameters found, given the data available**
- The likelihood of an example
- The support of a class
- The likelihood of an attribute, given the class label

**67) What is the difference between classification and regression?**

- **Classification has a categorical target, while regression has a numeric target**
- Classification is a supervised activity. while regression is unsupervised
- Classification can make errors, while regression is always exact
- Classification can have a numeric or categorical target, while regression has always a categorical target

**68) When is polynomial regression appropriate?**

- **When the relationship between the predicting variable and the target cannot be approximated as linear**
- When the target values are not linearly separable
- When there is more than one predicting attribute
- When it is necessary to project the data into a dimensional space

**69) How is the number of clusters determined in agglomerative clustering?**

- By assigning each data point to a cluster based on the final structure obtained from the dendrogram.
- **By cutting the dendrogram at a certain height.**
- By merging the closest clusters according to the defined distance metric.
- By computing the pairwise similarity or distance between all clusters.

**70) In data preprocessing, which of the operations below performs aggregation?**

- Combination of two attributes in order to obtain a more general attribute with smaller variability
- **Reduction of the number of distinct values of an attribute. in order to reduce its variability**
- **Reduction of the number of rows of the dataset by applying a grouping on one or more attributes and computing the measures as the results of aggregation functions**
- Combination of two or more datasets in order to obtain more information

**71) Match specific situations of data to the most appropriate pre-processing activity**

- The numeric attributes have very diverse ranges of values : **Rescaling**
- An attribute has a very skewed distribution (e.g. most of the values concentrated in a small range, and a very long tail of outliers): **Non linear transformations**
- A categorical attribute has a too large number of distinct values: **Aggregation**
- A nominal attribute needs to be used as predicting when using the sklearn library: **OneHot Encoding**

**72) What is the first step in agglomerative clustering? (NON VERIFICATA)**

Scegli un'alternativa:

- Cutting the Dendrogram: Decide on the number of clusters by cutting the dendrogram at a certain height
- Pairwise Cluster Similarity Calculation: Compute the pairwise similarity or distance between all clusters
- **Initialization: Start with each data point as a singleton cluster**
- Cluster Assignment: Assign each data point to a cluster based on the final structure obtained from the dendrogram

**73) Which of the following situations could suggest the use of MinMax scaling? (also known as "rescaling")**

- **There are numeric attributes with very diverse value ranges**
- There are attributes with very skewed data distributions
- There are attributes with a very large number of missing values
- There is a numeric attribute with values spread in a very large range